

NGUYỄN GIA NGHI

Nicholas Nguyen | Academic CV | Updated August 2026

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RESEARCH INTERESTS

Computer systems performance | Distributed systems & coordination | Software engineering

Latency diagnosis, Internet multimedia & dependable edge services

EDUCATION

VNUHCM-University of Science

Sep 2023–May 2027

Bachelor of Science | Major: Information Technology

Ho Chi Minh City

GPA: 8.34/10.00 (3.58/4.00) | 121 credits completed

Expected thesis completion: March 2027

UNDERGRADUATE RESEARCH THESIS

Latency diagnosis and optimization mechanisms for edge cloud-gaming systems | In progress

Supervisor: Professor Lê Ngọc Sơn | **Official title (Vietnamese):** Nghiên cứu cơ chế chẩn đoán và tối ưu hóa độ trễ cho hệ thống trò chơi đám mây biên

- Investigating whether reusable, context-specific latency fingerprints, stored patterns of system response keyed by node, workload, encoder, network, and client, can distinguish cross-layer bottlenecks and guide cause-specific recovery with less quality loss than generic adaptation.
- Built an operational, self-hosted edge-cloud-gaming testbed with a desktop-orchestrated, Dockerized game/runtime host, GStreamer capture and encoding, WebRTC media and input paths, a lightweight browser client, and cross-layer telemetry.
- Developing versioned observation and fingerprint schemas, a persistent fingerprint store, an analytical matcher that reports confidence and mixed/unknown states, and a bounded-probe planner that gathers additional response evidence only when passive observations remain ambiguous.
- Planned evaluation compares static, threshold-based, and passive-telemetry baselines against fingerprint-guided control, with and without deadline-aware scheduling, using diagnosis accuracy, P95/P99 latency, frame-deadline misses, recovery time, quality degradation, and diagnostic-probe overhead.

RELEVANT ACADEMIC PREPARATION

Selected strong results (10-point scale): Discrete Mathematics (9.40) | Introduction to Cloud Computing (9.10) | Computer Systems (9.10) | Software Design (8.50) | Applied Statistics for Engineers and Scientists (8.10)

Operating Systems (8.00) | Introduction to Databases (8.00)

Additional core coursework: Introduction to Software Engineering | Computer Networks | Data Structures and Algorithms

SELECTED SYSTEMS & SOFTWARE PROJECTS

PIXELATED Studio Edition | Creator platform & self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into a self-hosted edge-cloud-gaming prototype and research testbed in which interactive workloads run on a user-owned host and stream to a lightweight browser client.
- Implemented a hosted Fastify control plane, Electron-based host orchestration, a Dockerized runtime, GStreamer capture and encoding, secure pairing and session flows, WebRTC media and input paths, and cross-layer telemetry.

[Repository](#) | [Live application](#) | [Releases](#)

PIXELATED User Edition | Browser-based WebAssembly gaming platform

- Built and deployed a React/PWA application that runs supported NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly-based libretro cores, without relying on the Studio desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser-execution baseline for comparison with Studio’s edge-cloud-streaming path.

[Repository](#) | [Live application](#)

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern - TMA Solutions

Jun–Aug 2026

Ho Chi Minh City

- Worked with Jira tickets spanning bug fixes, product improvements, and small features across an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Modeled and queried MongoDB data through Mongoose, built API tests, self-tested changes, and submitted completed work for senior-engineer code review within Agile/Scrum sprints.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud platforms	AWS, GCP
Infrastructure & DevOps	Linux, Docker, Kubernetes, Helm, Terraform, GitHub Actions
Real-time media	WebRTC, GStreamer
Performance evaluation	Telemetry instrumentation, benchmarking, experimental design, latency analysis

CERTIFICATIONS & ENGLISH

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall - Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2 October 2025)

RESEARCH OUTPUT STATUS

Undergraduate thesis and supporting research artifacts are in progress; expected thesis completion is March 2027.